2026-04-18 : Briefing Document

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Briefing on Classification Modeling, Evaluation Metrics, and Exploratory Data Analysis

Executive Summary

This briefing synthesizes key concepts from the April 18, 2026, session regarding the application of machine learning classification models, the nuances of model evaluation (specifically the detection of overfitting), and the structured approach to Exploratory Data Analysis (EDA). The central takeaway is that machine learning cannot be treated as a "black box"; success requires an iterative, cyclical approach—often following the CRISP-DM framework—where domain knowledge and rigorous statistical checks of data features are prioritized over simple accuracy scores. A critical warning is issued regarding "perfect" training parameters, which often signal overfitting rather than model excellence.

I. Mechanics of Classification and Model Selection

Classification aims to define boundaries between discrete classes within a dataset to minimize misclassification. In real-world scenarios, classes are rarely neatly separated and often intermingle, necessitating complex algorithms beyond simple linear models.

A. Algorithmic Boundaries

The shape of a class boundary is dictated by the specific algorithm and its parameters:

- **Logistic Regression:** Typically results in linear boundaries.
- **Support Vector Classifier (SVC):** Uses kernels to define boundaries. A linear kernel forces linear separation, while a **Radial Basis Function (RBF) kernel** allows for curvilinear boundaries.
- **Random Forest:** The trees in the Random Forest partition the space in alignment with the horizontal and vertical axes (in n-dimensions), and the final prediction is created by combining the predictions from the basic trees. Parameters like the minimum number of observations per leaf node significantly affect the "smoothness" of these partitions.
- **Neural Networks:** Use hidden layer architectures to learn complex boundaries, though they may not always yield a superior model compared to simpler ensemble methods like Random Forest depending on data nature.

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B. Evaluation Metrics

Relying solely on accuracy is discouraged in classification as it can be a poor indicator of performance, especially with imbalanced data. Instead, a multi-metric approach is required:

Metric	Purpose	Key Considerations
Precision	Measure of exactness.	High precision means few false positives.
Recall	Measure of completeness.	High recall means few false negatives.
F1 Score	Harmonic mean of Precision and Recall.	Recommended as the primary metric for final model decisions.
ROC/AUC	Area Under the Receiver Operating Characteristic curve.	Values closer to 1 indicate a sharp ability to differentiate between classes.

II. Analyzing Model Fit: Overfitting vs. Underfitting

A primary responsibility of the data scientist is to identify when a model has failed to generalize.

A. Overfitting

Overfitting occurs when a model is too influenced by the training data, capturing noise rather than the broad trend. This happens when the model is overly flexible (Eg. too many polynomial features included in the feature set)

- **The "Gap" Indicator:** A "huge gap" between training metrics (e.g., perfect 1.0 precision) and testing metrics (e.g., 0.78 precision) is a red flag.
- **Visual Evidence:** In Random Forests, allowing only one observation per leaf node often leads to sharper, overly complex boundaries that perform perfectly on training data but fail on test data.
- **Rule of Thumb:** "Perfect parameters when you are doing machine learning are always suspect."

B. Underfitting

Underfitting occurs when the model is unreasonably rigid and fails to capture the underlying structure.

- **Indicator:** Both training and testing errors remain high and values are poor across the board.

III. The Data Science Lifecycle (CRISP-DM)

The session emphasizes the **Cross-Industry Standard Process for Data Mining (CRISP-DM)**, which prescribes six cyclical steps.

1. **Business Understanding:** Defining goals and how to evaluate success. This is critical for budgeting and management alignment (e.g., predicting employee retention to reduce costs).
2. **Data Understanding:** Initial investigations to gain insights and spot anomalies (EDA).

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3. **Data Preparation:** Transformations, feature engineering, and cleaning (80-90% of project time).
4. **Modeling:** Selecting and applying ML algorithms.
5. **Evaluation:** Analyzing errors and determining if goals are reached.
6. **Deployment:** Integrating the model into IT infrastructure (Cloud, VMs, etc.).

Note on Iteration: The process is cyclical. If error analysis reveals missing details, practitioners must backtrack from Modeling or Evaluation to Data Understanding.

IV. Structured Exploratory Data Analysis (EDA)

EDA is the approach used to summarize dataset characteristics and discover patterns. The session provides a "mind map" to categorize common data problems.

A. Within-Column Problems (X-Variables)

These issues occur within a single independent variable:

- **Missing Values/Incorrect Data:** Gaps in data or values that deviate significantly (outliers).
- **Incorrect Representation:** Using numbers for nominal data (like Gender) in a way that implies an unintended mathematical hierarchy (e.g., F=1 > M=0).
- **Statistical Distribution:** Some models (e.g., Gaussian) assume normal distribution. Non-normal data must be identified via KDE plots or histograms.
- **Heteroscedasticity:** Occurs when the variance of data changes over time (e.g., the "spread" increases). Remedies include Log or Box-Cox transformations.

B. Across-Column Problems (X-Variables)

These issues emerge from the relationships between independent variables:

- **Insufficient Features:** The relationship might be quadratic/cubic, but only one linear variable is provided. This requires **Polynomial Regression** or domain-specific feature engineering.
- **Too Many Features:** Leads to the "curse of dimensionality," requiring reduction techniques like **Principal Component Analysis (PCA)**.
- **Scaling Problems:** When variables have vastly different magnitudes (e.g., one variable ranges from 4-5 while another ranges from 40,000-50,000). Models like linear regression can be overwhelmed by the larger values, necessitating feature scaling.
- **Multicollinearity:** When one column can be predicted by a combination of others. This is identified through **Variance Inflation Factor (VIF)** analysis.

C. Dependent Variable (Y) Problems

- **Missing Labels:** If the class (Y) is not provided, unsupervised methods like **Clustering** can be used to discover classes before supervised learning can occur.

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V. Key Institutional Directives

The instructor emphasized the following regarding the course and professional development:

- **Mandatory Exercises:** Exercises are not just for practice but for building intuition.
- **Penalty Policy:** A **penalty of 2 marks** is imposed for failure to submit exercises. This serves as an "incentive" to ensure students score at least the 20-mark minimum required to pass the 50-mark end-semester examination.
- **Mathematical Depth:** This semester focuses on the lifecycle and process; the deep mathematics and intricacies of hyperparameter tuning for algorithms will be covered in the subsequent semester.
- **Model Persistence:** In professional deployment, models are "saved" by storing their coefficients (e.g., in **Pickle files**) so they can be read and used for prediction later without retraining.

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Study Guide: Data Science Lifecycle and Exploratory Data Analysis

This study guide provides a comprehensive review of the concepts discussed in the transcript dated April 18, 2026, focusing on the evaluation of classification models, the Data Science Lifecycle (CRISP-DM), and the foundational principles of Exploratory Data Analysis (EDA).

Part 1: Short-Answer Quiz

Instructions: Answer the following ten questions in 2–3 sentences each.

1. What determines whether a machine learning problem is categorized as regression or classification?
2. How does a classification algorithm handle real-life data where classes often overlap?
3. Explain the significance of the "cyclical" nature of the CRISP-DM process.
4. What is "overfitting," and how can it be identified using training and testing metrics?
5. Why is the "Business Understanding" phase placed first in the CRISP-DM framework?
6. What is the "curse of dimensionality," and how does it relate to feature selection?
7. Describe the difference between a linear kernel and a radial kernel in Support Vector Classifiers (SVC).
8. What is "heteroscedasticity," and which transformations can be used to address it?
9. Why is the F1 score considered a more reliable metric than accuracy in many classification scenarios?
10. What are the consequences of failing to submit course exercises according to the faculty's policy in this course?

Part 2: Quiz Answer Key

Question	Answer
1	The categorization depends on the nature of the dependent variable (Y); if Y is discrete, the problem is classification, and if Y is continuous, it is regression. This distinction usually dictates the choice of algorithms and evaluation metrics.
2	In real-life data, classes are rarely neatly separated, leading algorithms to focus on minimizing misclassification rather than achieving perfect separation. This results in the creation of boundaries (linear or curvilinear) where some observations are inevitably misclassified, which is reflected in precision and recall metrics.
3	CRISP-DM is cyclical because practitioners often discover missing details or errors during the modeling or evaluation stages that require backtracking. For example, poor model results may necessitate a return to the data understanding phase to re-evaluate assumptions or parameters.

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4	Overfitting occurs when a model becomes too influenced by the training data, following individual points rather than broad trends. It is identified by a large gap between performance metrics, where the model shows very low error on training data but high error on test data.
5	Business Understanding is the first step because it establishes the problem's goals, success criteria, and justifications for budgeting and effort. Linking machine learning models to financial or operational objectives is essential for convincing management and ensuring the project delivers value.
6	The curse of dimensionality refers to the problems created by having too many features (columns) in a dataset, which can overwhelm an algorithm. To fix this, practitioners use dimensionality reduction techniques (like Principal Component Analysis (PCA)) to remove unnecessary or redundant data.
7	A linear kernel forces the classification boundary to be a straight line, which is characteristic of models like linear kernel based Support Vector Classification. A radial kernel (such as a Radial Basis Function) allows for curvilinear boundaries, enabling the model to better navigate complex, overlapping data clusters.
8	Heteroscedasticity occurs when the variance of the data is not constant, often increasing over time or across a range of values. Remedies include applying log transformations or Box-Cox transformations to reduce variance and satisfy the assumptions of algorithms like linear regression.
9	Accuracy is a poor indicator when classes are imbalanced, as it can hide the poor performance of a model on minority classes. The F1 score, the harmonic mean of precision and recall, provides a more balanced view of how well the model differentiates between individual classes.
10	There is a specific penalty of 2 marks for students who do not submit their exercises. These assignments are critical for building the necessary skills.

Part 3: Essay Questions

Instructions: Use the provided source context to develop comprehensive responses to the following prompts.

1. **The Role of EDA in the Data Science Lifecycle:** Analyze why Exploratory Data Analysis and Data Preparation typically consume 80–90% of a data scientist's time. Discuss the specific anomalies and characteristics EDA aims to uncover.
2. **Evaluating Classification Models:** Compare and contrast the boundary-learning behaviors of Logistic Regression, Random Forest (with varying leaf node parameters), and Neural Networks. How do these behaviors impact the model's ability to generalize to new data?
3. **The EDA Mind Map Framework:** Explain the "within-column" versus "across-column" structural approach to identifying data problems. Provide examples of issues found in each category, such as missing values and multicollinearity.

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4. **Feature Engineering and Scaling:** Discuss the necessity of feature engineering and scaling. Explain how a "giant" variable with a large magnitude (e.g., 40,000) can overwhelm a smaller variable (e.g., 4.0) and how this affects the optimization process of machine learning models.
5. **Data Fidelity and Interpretation:** Address the faculty's warning that "perfect parameters are always suspect" and that "machine learning cannot be treated as a black box." Explain how domain knowledge interacts with mathematical metrics to produce a reliable model.

Part 4: Glossary of Key Terms

- **Accuracy:** A classification metric measuring the ratio of correct predictions to total observations; often considered a poor indicator in imbalanced datasets.
- **Box-Cox Transformation:** A mathematical transformation used to stabilize variance and make data more closely resemble a normal distribution.
- **Classification:** A supervised learning task where the target variable is discrete or categorical.
- **Collinearity/Multicollinearity:** A situation where one independent variable can be predicted by a combination of other independent variables, often measured via Variance Inflation Factor (VIF).
- **CRISP-DM:** Cross-Industry Standard Process for Data Mining; a six-step cyclical framework for data science projects.
- **Dependent Variable (Y):** The variable being predicted or explained in a machine learning model.
- **Exploratory Data Analysis (EDA):** An approach to analyzing datasets to summarize their main characteristics, discover patterns, and spot anomalies, often using statistics and visualization.
- **F1 Score:** The harmonic mean of precision and recall, used to evaluate the balance between the two metrics.
- **Feature Scaling:** The process of bringing all independent variables to a similar range or level to prevent variables with larger magnitudes from dominating the model.
- **Heteroscedasticity:** A condition in which the variance of the error terms or the data itself is not constant.
- **Independent Variable (X):** Also known as a "feature"; a variable used to predict the value of the dependent variable.
- **Overfitting:** When a model is too closely tailored to the training data, resulting in poor performance on unseen test data.
- **Pickle File:** A standard mechanism used to save a trained machine learning model's coefficients and state for future deployment.
- **Precision:** A metric indicating the accuracy of positive predictions made by the model.
- **Recall:** A metric indicating the model's ability to identify all relevant instances within a specific class.

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- **Regression:** A supervised learning task where the target variable is continuous.
- **Underfitting:** A situation where a model is too rigid and fails to capture the underlying trend of the data, resulting in high error for both training and testing sets.

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Here is a table of the questions asked during the sessions along with their corresponding answers, followed by a list of questions that were deferred or not answered.

Answered Questions

Question	Answer
"What is the difference between accuracy and precision?"	Accuracy refers to how close a result is to the true or correct value, or the proportion of all correct predictions. Precision refers to how consistent/repeatable the results are, or how many predicted positives were actually positive.
"If 2 or more than 2 models are performing well and almost same on train and test data, how do we choose the better one?"	If metrics are identical, you can conclude both methods perform equally well. However, you can decide by choosing the simpler model (fewer parameters), or by considering efficiency (speed, memory) and deployment constraints.
"Assignment number 8, there are four questions. So, the observation is supposed to be shared... how do we share it? ...IPVYNB files and zip it... Or can we document it in a document and then upload it?"	The faculty suggested creating a PDF with the observations, but stated they would check with the TA group and clarify it later that day.
"Which one usually we choose, even though test is performing there well, but the difference is huge with the train?"	If there is a huge gap between training and testing data, the model has likely overfit and performance is suspect. You should redo the model instead of relying on it.
"I just wanted to know if there are 3 parameters. What would happen to the display of scatter plots, sir?"	You can choose dominant variables to plot against, or use techniques like pair-plots, or TSNE (which will be learned later) to convert n-dimensional data into two-dimensional data for visualization.
"Depending upon your output that you are predicting, you are classifying into regression or classification. Can I go like that?"	Yes. If the dependent variable (Y) is discrete, it is a classification problem. If Y is continuous, it is a regression problem. These are the broad guidelines.
"There was a table for model and precision recall, so there were two, three columns of recall 1, recall 2. So, what are those?"	Because there are multiple classes (e.g., 4 classes), the metrics are derived and evaluated for each individual class.
"In regression also, we did random forest, and in classification also, we did random forest. (Implied: How is it used for both?)"	Many ML algorithms, like RandomForest, can be used for both. In regression, the algorithm predicts the observations (often visualized in the form of a

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	curve). In classification, the model effectively creates boundaries between different classes.
"Is underperformance of test data compared to train data an overfit in classification?"	Yes. A large difference where test errors are significantly higher than training errors is indicative of an overfit model.
"Where the test is predictable better than the train. So, is this a case of underfit?"	No, that usually happens by chance, or it may be attributable to data quality. In a true case of underfit, both the training and testing errors will be very high because the model is unreasonably rigid.
"Can we sort of formulate composite metric which can combine the test metrics, and the train metrics, and across different classes, and just give you one parameter that you can make a decision on?"	Trying to hide or over-abstract data can lead to erroneous decisions. You should visualize and process the matrices comprehensively using programming (e.g., Pandas) rather than relying on a single heavily abstracted number.
"Can we sort of tweak at least that [probability] selection so that test and train assessment or, modeling can be optimized?"	Yes, modeling functions provide parameters that allow you to set or change the probability threshold based on your specific needs (e.g., medical tests requiring over-safety).
"If I had to pick between the two [models with similar scores]... should I go with precision, or should I rely on the other parameters?"	You can be guided by metrics like F1 score, which is a harmonic mean and good combination of both precision and recall. However, you must take a good look at all fundamental parameters in totality rather than relying on just one.
"If we had, 30, 40 classes, should we approach the problem same way, or... Do we do something differently?"	The nature of the problem is the same, but visualizing it becomes difficult. You will have to rely on programmatic numeric evaluations (minimums, maximums, averages) rather than visual scatter plots.
"Can [cross validation] techniques be applied here as well in classification problems?"	Yes, cross-validation and regularization are common methods applicable to all of these algorithms.
"When we say feature, so we are actually talking about the column in the data, or more than that?"	Broadly, yes. Features refer to the independent X variables (columns), but they represent a domain-specific characteristic rather than just a number.
"Do we have any process that we, clean the data first... and based on that result, model can decide whether now the corrected data should be picked or not?"	There are no golden rules, but cleaning is a core part of Exploratory Data Analysis (EDA). As per the CRISP-DM standard, it is an iterative cycle; if model errors are high, you return to EDA to investigate and clean missing or duplicate values.

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"Is there a way that a model can be continuously trained... after the program terminates?"	This is the focus of a set of activities known as MLOps. It involves continuous tracking of inputs, outputs, monitoring errors, etc. to decide if the models need to change. (In the class the question was misunderstood, and the answer given – pickle files - pertained to re-use of models created during training)
"How do these large language models work? ... How do they know that something is incorrect?"	Before deployment, developers perform extensive error analysis by comparing LLM outputs to known correct results.
"Can we give some wrong feedback and make the [LLM] model incorrect? We can hack the models?"	If a model strictly adapted to wrong feedback, it would generate wrong results. However, LLMs have safeguards and their own mechanisms to verify correctness rather than relying entirely on user feedback.
"What are the ways that we can... identify that though those parameters are correct [high mean, high R-square], but it is not the analysis that the model is a good fit [due to overfitting]?"	Math is only a guideline. You must finally rely on domain knowledge to judge whether the model's outputs and acceptable error margins actually solve the practical problem in your specific field.
"Do we get any lecture to understand the basics of what [random forest or support vector algorithms] are doing?"	The mathematical intricacies of these algorithms will be covered deeply in courses during the next semester, as this current course focuses broadly on the entire data science life cycle.

Unanswered or Deferred Questions

The following questions were asked but were explicitly deferred to a later time or could not be answered with the given context:

- **"Can we do feature engineering?"** (The faculty deferred this by stating, *"We'll come to feature engineering later on."*)
- **"When do we actually have to look at this [CAUC] option, sir?"** (The faculty deferred answering immediately, noting that AUC requires deeper discussion and they would provide notes to discuss it later.)
- **"If we have to compare the ROC cut for multiple models or two models, we can do that, sir?"** (The faculty deferred this, again stating that ROC is a deeper topic that will be addressed with notes and discussion later.)
- **"In this [hypothetical overfitting] scenario which I explained, will there be a issue with, like, the random errors? Like, will the, errors not be random?"** (The faculty could not answer this without seeing a concrete dataset, stating *"I don't know. See, the point is, if you come to me with a data set, then we can talk something concrete."*).

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